

PLASMA SPRAY PROCESSORS

Evaluation of Thermal Spray Coating Procedure

QF-QMS/02/41

Rev. 00

Date : 12.03.2025



1. PURPOSE

To outline the procedure for evaluating the quality and conformance of thermal spray coatings through standardized testing methods.

2. SCOPE

Applicable to all thermal spray-coated components and test coupons prepared in the facility.

3. REFERENCES

- ASTM C633 – Adhesion Test
- ASTM D7091 / ISO 2178 – Coating Thickness
- ASTM E384 – Microhardness Test
- ASTM E18 - Macrohardness Test
- ASTM E2109 – Porosity Measurement
- ASTM E1920 – Microstructure Evaluation
- ASTM D4417 – Surface Roughness
- ASTM D522 – Bend Test

4. RESPONSIBILITIES

- Lab Technician: Perform and document all tests.
- QC Engineer: Verify results and ensure compliance.
- QA Head: Approve final reports.

5. EQUIPMENT REQUIRED

- Surface Roughness Tester / Testex Tape
- Coating Thickness Gauge (Eddy Current/Magnetic)
- Pull-Off Adhesion Tester
- Metallurgical Microscope
- Microhardness Tester
- Sample Preparation Tools (Cutter, Grinder, Polisher)

6. SAMPLE PREPARATION

- Select coated test coupons or component sections.
- Mount, grind, and polish to mirror finish.
- Etch for microstructural evaluation if required.

7. TESTING PROCEDURE

7.1 Coating Thickness

- Standard: ASTM D7091
- Procedure:
 - Calibrate instrument.
 - Take 5 readings.
 - Record average, min, and max.

7.2 Surface Roughness (Pre-Spray)

Prepared BY	Laxmi Patil	Signature		Date	
Verified By	Prasanna Wadkar	Signature		Date	

PLASMA SPRAY PROCESSORS

Evaluation of Thermal Spray Coating Procedure

QF-QMS/02/41

Rev. 00

Date : 12.03.2025



- Standard: ASTM D4417
- Procedure:
 - Measure Ra or use Testex tape.
 - Record values before coating.

7.3 Adhesion Test

- Standard: ASTM C633
- Procedure:
 - Bond specimen with epoxy.
 - Cure and test pull-off strength.
 - Record force and failure mode.

7.4 Microstructure Evaluation

- Standard: ASTM E1920
- Procedure:
 - Use microscope to inspect bond line, Interfacial contamination, oxides.
 - Capture 20 images of each.

7.5 Microhardness Test

- Standard: ASTM E384
- Procedure:
 - Perform indentations across coating.
 - Take 5 readings.
 - Record average hardness.

7.6 Bend Test

- Standard: ASTM D522
- Procedure:
 - Perform Bend on 15/25mm dia rod and bend to 90/180 degree across coating
 - Check for crack.
 - No spalling/lifting acceptable, Coating spalling/lifting unacceptable.

7.7 Porosity Measurement

- Standard: ASTM E2109
- Procedure:
 - Use image analysis or manual grid.
 - Capture 20 images.
 - Report porosity %

8. REPORTING

- Job/Coupon No.: _____
- Coating Type: _____
- Surface Roughness (Ra): _____ μm
- Coating Thickness: Avg _____ μm (Min: ____, Max: ____)
- Adhesion Strength: _____ MPa (Perform in another Lab)
- Failure Mode: ☐ Cohesive ☐ Adhesive ☐ Substrate
- Microstructure Observations:
- Porosity: _____ %
- Hardness: _____ HV0.3/HRC/HR15N/HR15Y

Page 2 of 3

Prepared BY	Laxmi Patil	Signature		Date	
Verified By	Prasanna Wadkar	Signature		Date	

PLASMA SPRAY PROCESSORS

Evaluation of Thermal Spray Coating Procedure

QF-QMS/02/41

Rev. 00

Date : 12.03.2025



- Bend Test: Acceptable/Unacceptable
- Final Status: ☐ Pass ☐ Fail

9. SAFETY PRECAUTIONS

- Always wear lab PPE.
- Use gloves when handling chemicals.
- Operate machinery with guards in place.

10. DOCUMENT CONTROL

All test records and reports shall be maintained for a minimum of 5 years or as per quality system requirements.

Prepared BY	Laxmi Patil	Signature		Date	
Verified By	Prasanna Wadkar	Signature		Date	